Ryan R. Hu

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Education

California Institute of Technology (Caltech) Pasadena, CA
B.S. Computer Science - *GPA: 4.2/4.0* *Graduation Date: Jun 2027*

Relevant Coursework: Software Design, Deep Learning, Learning Systems, Machine Learning & Data Mining, GPU Programming, Algorithms, Computing Systems, Diffusion Models, Differential Equations, Probability & Statistics, Probability Models

Teaching Assistant: Decidability and Tractability (Jan 2025 - Mar 2025), Machine Learning & Data Mining (Jan 2026 - Mar 2026)

Technical Skills

Languages: Python, Java, C, C++, JavaScript/TypeScript, HTML/CSS, SQL, MATLAB, LaTeX
Technologies/Frameworks: Git, React.js, Node.js, Pandas, OpenCV, PyTorch, Keras, TensorFlow, Scikit-Learn, Tailwind, Flask, Plotly, PySpark, SLURM, High Performance Computing (HPC), Jupyter
Skills: CI/CD, Machine Learning, Full Stack, Data Analysis, Debug Logging, Containerization, Unit Testing

Experience

Amazon - Business Payments Team Seattle, WA
Software Development Engineer Intern *June 2026 – Present*
- Incoming Summer 2026

Tracevision - Math Team Los Angeles, CA
Software Engineering & Machine Learning Intern *Jun 2025 – Sep 2025*
- Automated data preprocessing pipelines and scripts to extract and construct 6000+ shot sequences from 80+ hours of raw basketball game footage centered around the basket, optimizing video format and quality for annotation teams through frame-level segmentation and noise reduction.
- Designed and deployed production-grade hybrid CNN + Transformer shot detection model, leveraging MobileNetV2 as the CNN backbone, achieving 95% accuracy on made vs. missed shot classification from annotated video clips.
- Improved ptools, a full-stack internal tool for object tracking and annotation review, by optimizing components to boost efficiency and enhance usability for 100+ users - cutting detection load latency from 1.2s to 0.7s ($\sim$ 40% faster). Built with Python, MySQL, HTML/CSS, and JavaScript.
- Developed geospatial API deployment pipelines using a combination of custom scripts and AWS infrastructure (EC2, S3, Lambda), streamlining integration workflows for 30+ facilities and 100+ cameras in security and retail sectors.

Caltech Pasadena, CA
Undergraduate Researcher - Wierman Lab - Advised by Dr. Christopher Yeh and Dr. Adam Wierman *Apr 2025 – Present*
- Researching diffusion-based approaches for constrained optimal power flow in energy system contexts.

Undergraduate Researcher - Alvarez Lab - Advised by Dr. Rafal Kocielnik and Dr. R. Michael Alvarez *Feb 2025 – Present*
- Building custom autoencoder architecture to detect anomalous player behavior in Activision's Call of Duty (COD) games, using SQL and Python to clean and process 3M+ raw gameplay logs, identifying 1000+ anomalous players within a 10-day window through reconstruction error.
- Designing a verification pipeline integrating UMAP + HDBSCAN clustering with temporal segmentation, identifying distinct types of anomalous behaviors and analyzing their impact on player engagement in 20,000+ players.

Undergraduate Researcher - Alvarez Lab - Advised by Dr. Cong Cao and Dr. R. Michael Alvarez *Apr 2024 – Dec 2024*
- Analyzed HIV patient datasets collected by the Multicenter AIDS Cohort Study (MACS) by applying causal ML and gradient boosting trees on 800+ patients, discovering significant epigenetic markers correlated with accelerated aging - currently authoring manuscripts on findings.
- Built data preprocessing pipeline merging epigenetic datasets with web-scraped environmental satellite data, performing data cleaning, imputation, and feature transformation for machine learning tasks downstream.

Undergraduate Researcher - GALCIT Lab - Advised by Dr. Roni Goldshmid and Dr. John O. Dabiri *June 2023 - Sep 2023*
- Developed a visual anemometry framework in MATLAB combining YOLOv3 object detection (95.4% precision) with optical flow mapping techniques to estimate wind speeds within ±5 m/s error.
- Conducted a quantitative analysis linking canopy motion variance to wind standard deviation across three different tree species, achieving $>$0.9 correlation in field tests.

Selected Projects

Neural Approximate Mirror Maps for DC-Optimal Power Flow | *Python, PyTorch*
- Developed a geometry-aware diffusion framework for DC Optimal Power Flow using Neural Approximate Mirror Maps (NAMMs) on the IEEE 118-bus benchmark, learning latent representations of feasible operating points.
- Reduced mean constraint violation by 92% (36.5 $\rightarrow$ 2.82) and generator-dispatch MAE by 40% (116.3 MW $\rightarrow$ 70.2 MW), increasing feasible samples with $<$100 MW error from 57.4% to 86.7%.

Voxel Rasterizer | *C++, SDL, CUDA*
- Developed a GPU-accelerated multi-view voxel reconstruction and rasterization pipeline, implementing shape-from-silhouette 3D
- Optimized CUDA kernels for occupancy and memory throughput, achieving 46.9$\times$ faster reconstruction and 32.9$\times$ end-to-end pipeline speedup over CPU baselines.

TinyLM | *Python, FastAPI, React, PyTorch*
- Implemented a lightweight transformer-based language model ($\sim$22M total parameters) trained on TinyStories dataset. Includes custom BPE tokenization, dataset pipeline, training loop, and autoregressive text generation. Exposed inference through FastAPI backend for real-time prompting and querying.
- Designed a full-stack web interface with React connected to the FastAPI backend, enabling prompt-based interaction and live text generation.

Agentic Stock Trader | *Python, LangChain, GCP, Gemini API*
- Built a multi-agent stock trading framework where LLM-driven agents ingest 1,000+ financial news items, debate with each other, and output explainable decisions, achieving up to 2.34% monthly return for various individual stocks in backtests.
- Designed a recursive agent debate architecture with static personas, dynamic leaf agents, and hierarchical clustering, creating 50+ distinct agents across 3 debate layers and multiple decision rounds to systematically evaluate trades and reach agreement.

Leadership/Activities

Caltech Senior Class Co-President Jun 2026 – Present
Hacktech Organizer - Caltech's Hackathon Dec 2025 – Present
NCAA Division III Men's Basketball Sep 2023 - Mar 2025